CB400 Molded Case Circuit Breaker

Technical Datasheet & Installation Manual

Model: CB400-3P-250A

Manufacturer: ElectroPower Systems

Category: Electrical

ETIM Code: EC000082

Document Number: DS-CB400-EN-RevB

Revision: B | Date: January 2025

Classification: Public

TABLE OF CONTENTS

1 PRODUCT OVERVIEW

1.1 Description

The CB400 is an IEC 60947-2 compliant MCCB for protection of electrical circuits against overload and short-circuit currents. Features thermal-magnetic trip unit with adjustable Ir and fixed Im settings, front-accessible wiring, and optional motorized operator for remote switching.

The CB400-3P-250A is engineered to meet the demanding requirements of modern industrial processes. Built by ElectroPower Systems, this product represents years of engineering expertise and rigorous testing to ensure reliable operation in harsh industrial environments. The device features state-of-the-art technology combined with robust construction to deliver exceptional performance and long service life.

1.2 Key Features

- High accuracy and repeatability for demanding applications
- Robust construction for harsh industrial environments
- Wide operating temperature range
- Multiple output options and communication protocols
- Easy installation and commissioning
- Low maintenance requirements
- Comprehensive diagnostics and self-monitoring
- Compact design for space-constrained installations
- Full compliance with international standards (IEC 60947-2, CE, UL 489)
- Long operational lifetime and warranty coverage

1.3 Applications

The CB400 Molded Case Circuit Breaker is designed for use in a wide range of industrial applications including:

- Chemical and petrochemical processing plants
- Oil and gas production and refining
- Water and wastewater treatment facilities
- Food and beverage manufacturing
- Pharmaceutical and biotech production
- Power generation and distribution
- Pulp and paper mills
- Mining and mineral processing
- HVAC and building automation systems
- Semiconductor and electronics manufacturing
- Automotive and general manufacturing
- Marine and offshore installations

1.4 Principle of Operation

The CB400-3P-250A operates using advanced sensing and control technology. The primary sensing element converts the physical measurement into an electrical signal which is then processed by the integrated microprocessor. The digital signal processing provides linearization, temperature compensation, and filtering to ensure accurate and stable output. The processed signal is then converted to the selected output format (analog, digital, or fieldbus) for transmission to the control system. Comprehensive self-diagnostics continuously monitor the device health and alert operators to any abnormal conditions.

2 TECHNICAL SPECIFICATIONS

2.1 General Specifications

Frame Size	400 A
Rated Current (In)	250 A (adjustable 200-250 A)
Poles	3-pole
Breaking Capacity (Icu)	50 kA at 415 V AC
Service Breaking (Ics)	100% Icu
Trip Unit	Thermal-Magnetic, adjustable
Utilization Category	A (non-selective)

2.2 Additional Parameters

Standards Compliance	IEC, EN, ANSI (Electrical specific)
Quality System	ISO 9001:2015 certified production
Country of Origin	Germany / USA
Typical Service Life	> 20 years under normal conditions
Mean Time Between Failures	> 150,000 hours
Warranty Period	3 years from date of delivery
Recommended Calibration	Every 12-24 months
Software Version	Firmware V3.2.1

2.3 Specification Summary Table

Parameter	Min	Typical	Max	Unit
230 V AC	65	65	---	
415 V AC	50	50	---	
500 V AC	36	36	---	
690 V AC	20	20	---	

3 ELECTRICAL RATINGS & WIRING

3.1 Electrical Specifications

Rated Voltage (Ue)	690 V AC
Insulation Voltage (Ui)	800 V
Impulse Voltage (Uimp)	8 kV
Frequency	50/60 Hz
Auxiliary Contacts	1 NO + 1 NC (standard)

3.2 Electrical Safety

All electrical connections must be made by qualified personnel in accordance with local electrical codes and regulations. Ensure that the supply voltage matches the device rating before energizing. Use appropriate circuit protection (fuses/breakers) as specified in this document.

WARNING

DANGER: Risk of electric shock. Disconnect all power sources before performing any wiring or maintenance. Allow a minimum of 5 minutes for capacitors to discharge after power is removed. Verify absence of voltage with a suitable measuring instrument before touching any conductors.

3.3 EMC Considerations

This product has been designed and tested to comply with applicable EMC standards. To maintain EMC performance in the installed system, observe the following guidelines:

- Use shielded cables for all signal connections, grounding the shield at one end
- Route signal cables separately from power cables (minimum 200 mm separation)
- Install appropriate surge protection devices on all field wiring
- Use twisted pair cables for analog signals to minimize interference
- Connect the device grounding terminal to the plant ground system
- Keep cable runs as short as possible
- Avoid routing cables near high-frequency switching devices (VFDs, contactors)

3.4 Wire Sizing Guide

Wire Size (mm2)	Wire Size (AWG)	Max Length (m)	Current Capacity (A)	Application
0.5	20	50	3	Signal / sensor
0.75	18	75	5	Low power
1.0	17	100	8	General
1.5	16	150	12	Power supply
2.5	14	200	16	Motor / heater
4.0	12	250	25	High power

4 MECHANICAL DIMENSIONS

4.1 Physical Dimensions

Width	140 mm (3P)
Height	260 mm
Depth	103 mm
Mounting	DIN rail / direct
Weight	3.8 kg

4.2 Dimensional Drawing Reference

All dimensions are in millimeters unless otherwise stated. Tolerances are per ISO 2768-mK. For detailed 2D/3D CAD models, please visit the ElectroPower Systems download portal or contact your local sales representative.

NOTE

CAD models available in STEP, IGES, and DXF formats at www.electropowersystems.com/downloads

4.3 Mounting Instructions

The device must be mounted in accordance with the following guidelines to ensure proper operation and to maintain the specified ingress protection rating:

1. Select a mounting location that provides adequate clearance for wiring and maintenance access.
2. Ensure the mounting surface is flat, clean, and capable of supporting the device weight plus dynamic loads.
3. Use the specified fasteners and torque values. Do not over-tighten as this may damage the housing.
4. Apply thread sealant (PTFE tape or liquid sealant) to all process connections as required.
5. Verify that the process connection type and rating match the installation requirements.
6. Orient the device as shown in the dimensional drawing for optimal performance.
7. Maintain the minimum clearances specified below for proper ventilation and heat dissipation.
8. Ensure all cable entries are properly sealed with the supplied cable glands.

4.4 Clearance Requirements

Direction	Minimum Clearance (mm)	Recommended Clearance (mm)	Purpose
Top	30	50	Ventilation / heat dissipation
Bottom	30	50	Cable routing
Left side	15	30	Adjacent device clearance
Right side	15	30	Adjacent device clearance
Front	50	100	Maintenance / display access

5 OPERATING CONDITIONS & ENVIRONMENT

5.1 Environmental Ratings

Ambient Temperature	-25 to 70 deg C
Storage Temperature	-40 to 85 deg C
Altitude	Up to 2000 m
Mechanical Life	20,000 operations
Electrical Life	10,000 operations

5.2 Derating Information

The device is rated for full performance within the specified operating temperature range. When operating outside the standard range, the following derating factors apply:

Ambient Temperature	Max Output	Accuracy Impact	Lifetime Factor
-40 to -20 C	90%	+0.1% FS	0.95x
-20 to 0 C	95%	+0.05% FS	1.0x
0 to 40 C	100%	Nominal	1.0x
40 to 60 C	95%	+0.05% FS	0.95x
60 to 70 C	85%	+0.1% FS	0.85x
70 to 85 C	75%	+0.2% FS	0.70x

5.3 Environmental Protection

The CB400-3P-250A is designed for installation in industrial environments. The enclosure provides protection against ingress of solid objects and water as specified by the IP rating. Additional protection measures may be required for installation in corrosive atmospheres, high-vibration environments, or areas subject to direct solar radiation.

- For outdoor installations, use a weather-protection enclosure or sun shield
- In corrosive environments, specify optional protective coatings
- For high-vibration areas, use vibration dampening mounts
- In wash-down areas, verify that the IP rating is sufficient for the cleaning procedures used
- For installations in classified hazardous areas, ensure compliance with applicable ATEX/IECEx requirements

6 MATERIALS OF CONSTRUCTION

6.1 Material Specifications

Housing	Glass-reinforced Polyamide PA6.6
Contacts	Silver alloy
Arc Chamber	Ceramic plates
Terminals	Tinned copper
Operating Mechanism	Toggle, stored energy

6.2 Chemical Compatibility

The wetted materials have been selected for broad chemical compatibility. However, the suitability of materials for specific process fluids must be verified by the end user. The following table provides general guidance on material compatibility with common process media:

Medium	316L SS	Hastelloy C	PTFE	FKM	EPDM	NBR
Water (neutral)	A	A	A	A	A	A
Sulfuric Acid (<50%)	B	A	A	B	B	C
Hydrochloric Acid	C	A	A	B	C	C
Sodium Hydroxide	A	A	A	A	A	A
Ammonia	A	A	A	C	A	C
Diesel / Fuel Oil	A	A	A	A	C	A
Acetic Acid	B	A	A	C	B	C
Chlorine (gas)	C	A	A	A	C	C

A = Excellent | B = Good (limited life) | C = Not recommended | Contact factory for specific media

6.3 Material Certificates

Material certificates (EN 10204 Type 3.1) are available for all wetted parts upon request. Please specify material certificate requirements at the time of ordering. Positive Material Identification (PMI) testing can be performed on request for critical applications.

7 PERFORMANCE DATA & CURVES

7.1 Performance Summary Table

Voltage	Icu (kA)	Ics (kA)	Icw (1s)
230 V AC	65	65	---
415 V AC	50	50	---
500 V AC	36	36	---
690 V AC	20	20	---

7.2 Accuracy & Error Budget

Error Source	Contribution (%FS)	Conditions	Notes
Non-linearity	0.05	BFSL method	Includes hysteresis
Hysteresis	0.02	Full range cycle	Max value
Repeatability	0.01	Same conditions	Short-term
Temp. effect (zero)	0.02/10K	Over comp. range	Per 10K change
Temp. effect (span)	0.02/10K	Over comp. range	Per 10K change
Long-term drift	0.1/year	Normal conditions	Annual maximum
Supply voltage effect	0.005/V	Within rated range	Per volt change
Total uncertainty	< 0.15	RSS method	95% confidence

7.3 Response Characteristics

The CB400-3P-250A response time is measured as the time required for the output to reach within a specified percentage of its final value following a step change in the input. The response time is affected by the damping setting, digital filter configuration, and the physical characteristics of the installation.

Damping Setting	Response Time (t63)	Response Time (t90)	Response Time (t99)	Update Rate
Off (fastest)	< 5 ms	< 12 ms	< 25 ms	100 Hz
Low	50 ms	115 ms	250 ms	20 Hz
Medium	200 ms	460 ms	1.0 s	5 Hz
High	1.0 s	2.3 s	5.0 s	1 Hz
Custom	Programmable	---	---	Variable

7.4 Long-term Stability Test Results

The following data represents typical long-term stability test results measured over a 12-month period at reference conditions (23 +/- 2 deg C, 45 +/- 10% RH):

Test Duration	Zero Drift (%FS)	Span Drift (%FS)	Total Drift (%FS)	Pass/Fail
1 month	< 0.01	< 0.01	< 0.02	PASS
3 months	< 0.02	< 0.02	< 0.04	PASS
6 months	< 0.04	< 0.03	< 0.07	PASS
12 months	< 0.06	< 0.05	< 0.10	PASS

8 INSTALLATION INSTRUCTIONS

WARNING

Read all instructions carefully before beginning installation. Improper installation may result in damage to the device, personal injury, or death. Installation must be performed by qualified personnel trained in the installation of industrial instrumentation and familiar with applicable safety regulations.

8.1 Pre-Installation Checklist

1. Verify that the received product matches the purchase order (model, range, options).
2. Inspect the device for any shipping damage. Report any damage to the carrier immediately.
3. Verify that the process conditions (pressure, temperature, media) are within the device ratings.
4. Ensure that all required tools, fittings, and accessories are available.
5. Review the installation location for accessibility, environmental conditions, and safety.
6. Verify that the electrical supply matches the device requirements.
7. Prepare all process connections with appropriate thread sealant.
8. Ensure the process is depressurized and drained before installation.

8.2 Mechanical Installation

Mount the CB400-3P-250A according to the dimensional drawing provided in Section 4. Follow these steps for proper mechanical installation:

1. Clean the process connection point thoroughly. Remove any burrs, debris, or old sealant material.
2. Apply 2-3 wraps of PTFE tape (or approved liquid sealant) to the male thread connection.
3. Hand-tighten the device into the process connection until finger-tight.
4. Use the appropriate wrench to tighten the connection to the specified torque value (see table below).
5. Do NOT use the device housing or electrical connector as a lever point during tightening.
6. Verify that the device is oriented correctly for the intended measurement.
7. Install any required mounting brackets or supports as needed.
8. Connect the impulse piping or process tubing if applicable.

8.3 Torque Specifications

Connection Type	Size	Torque (Nm)	Torque (ft-lbs)	Tool
NPT male	1/4 NPT	15-20	11-15	17 mm wrench
NPT male	1/2 NPT	25-35	18-26	22 mm wrench
G (BSP) male	G 1/4	20-25	15-18	19 mm wrench
G (BSP) male	G 1/2	30-40	22-30	27 mm wrench
Flange bolts	M12	60-80	44-59	19 mm socket
Flange bolts	M16	120-160	88-118	24 mm socket
Cable gland	M20	5-8	4-6	25 mm wrench

8.4 Electrical Installation

After completing the mechanical installation, proceed with the electrical connections:

1. Ensure all power sources are disconnected and locked out (LOTO procedure).
2. Remove the terminal cover or connector cap from the device.

3. Strip the cable conductors to the specified length (typically 7-10 mm).
4. Connect the wires according to the wiring diagram in Section 10.
5. Ensure cable shields are connected to the ground terminal (if applicable).
6. Tighten all terminal screws to 0.5-0.8 Nm (do not over-tighten).
7. Verify all connections with a continuity tester before applying power.
8. Install cable glands and seal all cable entries to maintain IP rating.
9. Replace the terminal cover and secure with the provided fastener.
10. Apply power and verify device operation according to Section 9.

9 COMMISSIONING & START-UP

9.1 Initial Power-Up

After completing the mechanical and electrical installation, follow this procedure for initial start-up of the CB400-3P-250A:

1. Apply power to the device. The display (if equipped) should illuminate within 2 seconds.
2. Observe the self-diagnostic sequence. The device performs a complete self-test during power-up.
3. Wait for the self-test to complete (approximately 5-10 seconds). A steady output indicates normal operation.
4. Verify the output signal is within the expected range using a calibrated multimeter or test instrument.
5. If the device has a display, verify that the reading corresponds to the actual process condition.
6. If the output or reading is not as expected, refer to Section 12 (Troubleshooting).

9.2 Zero and Span Adjustment

The device is factory-calibrated and should not require adjustment under normal circumstances. If field adjustment is necessary, follow this procedure:

1. Apply the zero reference condition (e.g., atmospheric pressure, 0 flow, ambient temperature).
2. Wait for the reading to stabilize (minimum 30 seconds).
3. Access the configuration menu: Press and hold [SET] button for 3 seconds.
4. Navigate to CALIBRATION > ZERO ADJUST.
5. Press [ENTER] to perform the zero adjustment.
6. Apply the span reference condition using a calibrated reference standard.
7. Navigate to CALIBRATION > SPAN ADJUST.
8. Press [ENTER] to perform the span adjustment.
9. Verify the calibration at 0%, 25%, 50%, 75%, and 100% of range.
10. Record the as-found and as-left calibration values.

9.3 Commissioning Checklist

Check Item	Requirement	Result	Initials
Power supply voltage	Within rated range	______	______
Output at zero	4.00 +/- 0.02 mA	______	______
Output at span	20.00 +/- 0.02 mA	______	______
Display reading	Matches reference	______	______
Alarm setpoints	Per specification	______	______
Communication	Response to polling	______	______
Leak test	No leaks at rated P	______	______
Documentation	As-built recorded	______	______

10 WIRING DIAGRAMS & CONNECTIONS

10.1 Terminal Assignment

Terminal	Function	Wire Color	Description
1 (+)	Supply / Signal +	Brown / Red	Positive supply and signal output
2 (-)	Supply / Signal -	Blue / Black	Negative supply / signal return
3	Output 2 / Alarm	White	Secondary output or alarm relay
4	Shield / Ground	Green/Yellow	Cable shield connection
5	Communication +	White/Green	RS485 / HART data (+)
6	Communication -	White/Brown	RS485 / HART data (-)

10.2 Connection Configurations

2-Wire Connection (4-20 mA loop powered):

Power Supply (+) ---- Terminal 1 (+) ---- Device ---- Terminal 2 (-) ---- Receiver ---- Power Supply (-)

4-Wire Connection (separate power and signal):

Power Supply (+) ---- Terminal 1 (+)
Power Supply (-) ---- Terminal 2 (-)
Signal Output ---- Terminal 3 ---- Receiver Input (+)
Signal Return ---- Terminal 4 ---- Receiver Input (-)

10.3 HART Communication Wiring

For HART communication, connect the HART communicator or modem across terminals 1 and 2. The minimum loop resistance for HART communication is 250 ohm. If the loop resistance is less than 250 ohm, install a 250 ohm resistor in series with the loop.

NOTE

HART communication requires a minimum loop impedance of 250 ohm at the HART signal frequency (1200/2200 Hz). Many modern transmitters include built-in loop resistance measurement - check the diagnostics menu.

11 CONFIGURATION & PROGRAMMING

11.1 Parameter Overview

Parameter	Range	Default	Access Level	Description
Unit	Bar/PSI/kPa/MPa	Bar	Basic	Engineering unit
Range Low	-100 to +99% FS	0.0	Basic	Lower range value
Range High	+1 to +100% FS	100.0	Basic	Upper range value
Damping	0.0 to 60.0 s	0.5 s	Basic	Output damping time
Output Mode	Linear/Sqrt/Custom	Linear	Advanced	Transfer function
Alarm Type	Hi/Lo/Window/Rate	Hi	Advanced	Alarm mode
Alarm Setpoint	Within range	90% FS	Advanced	Alarm trigger point
Fail-safe	Hi/Lo/Hold/Custom	Hi	Advanced	Output on fault
Display	Value/Bar/Both/%	Value	Basic	Display format
Protocol Address	0-247	1	Service	Modbus/HART address

11.2 Menu Structure

The device configuration is organized in the following menu hierarchy:

```
MAIN MENU
 |-- DISPLAY
 |    |-- Unit, Format, Orientation, Contrast
 |-- MEASUREMENT
 |    |-- Range, Damping, Filter, Output Mode
 |-- OUTPUT
 |    |-- Analog Output, Alarm, Simulation
 |-- COMMUNICATION
 |    |-- HART, Modbus, Device Tag, Polling Address
 |-- DIAGNOSTICS
 |    |-- Self Test, Peak Hold, Error Log, Runtime Counter
 |-- CALIBRATION
      |-- Zero, Span, Trim, Factory Reset
```

11.3 Configuration via HART

The device supports HART 7 protocol for remote configuration. Use any HART-compatible communicator or asset management software (e.g., PACTware, Endress+Hauser FieldCare, Emerson AMS) to access all device parameters remotely via the 4-20 mA loop.

12 MAINTENANCE & TROUBLESHOOTING

12.1 Preventive Maintenance Schedule

Task	Interval	Estimated Time	Skill Level	Parts Required
Visual inspection	Monthly	5 min	Operator	None
Verify output signal	Quarterly	15 min	Technician	Multimeter
Check connections	Semi-annual	15 min	Technician	Torque wrench
Calibration check	Annual	30 min	Specialist	Reference standard
Full calibration	2 years	60 min	Specialist	Reference standard
Seal / gasket check	2 years	30 min	Technician	Seal kit (optional)
Replace desiccant	Annual	5 min	Operator	Desiccant pack
Firmware update	As available	15 min	Specialist	Communicator / PC

12.2 Troubleshooting Guide

Symptom	Possible Cause	Corrective Action
No output	No power supply	Check supply voltage at terminals
No output	Wiring fault	Verify connections per wiring diagram
No output	Device fault	Check diagnostic LED / error code
Erratic reading	Electrical noise	Check cable routing and shielding
Erratic reading	Process turbulence	Increase damping setting
Erratic reading	Loose connection	Tighten all terminals
Offset error	Zero drift	Perform zero calibration
Span error	Span drift	Perform span calibration
Over-range	Process excursion	Verify process conditions
Over-range	Blocked line	Inspect impulse tubing
Communication fail	Wrong address	Verify protocol address setting
Communication fail	Baud rate mismatch	Match baud rate to host
High temperature alarm	Excess ambient temp	Improve ventilation or relocate
Sensor fault	Sensor damage	Replace device or sensor element

12.3 Error Codes

Error Code	Description	Severity	Action Required
E001	Sensor open circuit	Critical	Check sensor wiring, replace if needed
E002	Sensor short circuit	Critical	Check sensor wiring, replace if needed
E003	Over-range (high)	Warning	Verify process, adjust range if needed
E004	Under-range (low)	Warning	Verify process, adjust range if needed
E005	Temperature compensation error	Warning	Allow device to reach thermal equilibrium
E010	EEPROM write error	Critical	Cycle power, contact factory if persistent
E011	Calibration data invalid	Critical	Perform factory reset and recalibrate
E020	Communication timeout	Info	Check communication wiring and settings

13 SAFETY INFORMATION

WARNING

READ ALL SAFETY INFORMATION BEFORE INSTALLING, OPERATING, OR SERVICING THIS DEVICE. FAILURE TO FOLLOW THESE INSTRUCTIONS MAY RESULT IN DEATH, SERIOUS INJURY, OR PROPERTY DAMAGE.

13.1 General Safety Precautions

- This device must be installed, operated, and maintained by qualified personnel only.
- Follow all applicable local, national, and international safety codes and regulations.
- Always use appropriate personal protective equipment (PPE) when working on process equipment.
- Depressurize and drain the process before installing or removing the device.
- Disconnect and lock out all energy sources before performing any maintenance work.
- Do not exceed the rated pressure, temperature, or voltage specified in this document.
- Do not modify or alter the device in any way. Unauthorized modifications void the warranty and certifications.
- If the device is damaged or suspected of malfunction, remove it from service immediately.
- Dispose of the device in accordance with local environmental regulations at end of life.

13.2 Hazardous Area Safety

If this device is certified for use in hazardous (classified) areas (ATEX, IECEx, FM, CSA), the following additional precautions apply:

- Verify that the device certification matches the area classification before installation.
- Use only certified cable glands, conduit fittings, and accessories in the hazardous area.
- Do not open the device housing or disconnect wiring in a potentially explosive atmosphere.
- Maintain the specified maximum surface temperature (T-class) under all operating conditions.
- Refer to the device-specific ATEX/IECEx certificate for special conditions of use.

13.3 Pressure Equipment Safety (PED)

The CB400-3P-250A may be subject to the Pressure Equipment Directive (PED 2014/68/EU) depending on the process conditions and fluid group. Refer to the PED classification table provided with the device certification documentation. The user is responsible for ensuring compliance with PED requirements in the installed system.

14 ORDERING INFORMATION & ACCESSORIES

14.1 Model Number Structure

Model: CB400-3P-250A

The model number is structured as follows:
CB400 = Product family / base model
3P = Configuration / variant
250A = Material / option code

14.2 Available Options

Option Code	Description	Price Impact	Lead Time
STD	Standard configuration	Base price	2-4 weeks
-SS	316L stainless steel wetted parts	+15%	2-4 weeks
-HC	Hastelloy C-276 wetted parts	+45%	6-8 weeks
-TI	Titanium wetted parts	+65%	8-10 weeks
-HT	High temperature version (-40 to 200 C	+20%	4-6 weeks
-CT	Cryogenic version (-196 to 100 C)	+25%	6-8 weeks
-HART	HART 7 communication	+10%	Standard
-FF	Foundation Fieldbus	+20%	4-6 weeks
-PA	PROFIBUS PA	+20%	4-6 weeks
-ATEX	ATEX/IECEx Zone 0 certification	+15%	2-4 weeks
-SIL	SIL 2/3 certification	+20%	4-6 weeks
-CAL	5-point NIST traceable calibration	+5%	Standard

14.3 Recommended Spare Parts

Part Number	Description	Quantity	Recommended Stock
CB400-SEAL-01	Process seal kit (FKM)	1 set	1 per 5 devices
CB400-SEAL-02	Process seal kit (EPDM)	1 set	As required
CB400-GASK-01	Gasket set	1 set	1 per 5 devices
CB400-GLAND-01	Cable gland M20, nickel brass	1 pc	1 per 10 devices
CB400-CONN-01	Electrical connector, 4-pin M12	1 pc	1 per 10 devices
CB400-DESIC-01	Desiccant pack	5 pcs	Annual replacement
CB400-MOUNT-01	Mounting bracket set	1 set	As required

14.4 Recommended Accessories

Part Number	Description	Application
ACC-MNT-01	Pipe mounting kit (2 inch)	Pipe/pole mounting
ACC-MNT-02	Wall mounting bracket	Wall/panel mounting
ACC-SUN-01	Sun/weather protection shield	Outdoor installations
ACC-VLV-01	Manifold valve set (3-valve)	Isolation and calibration
ACC-VLV-02	Manifold valve set (5-valve)	Differential with equalize
ACC-CAB-05	Pre-made cable assembly (5 m)	Quick installation

ACC-CAB-10	Pre-made cable assembly (10 m)	Quick installation
ACC-DSP-01	Remote display unit	Display at distance
ACC-SPD-01	Surge protection device	Lightning protection

15 CERTIFICATIONS & COMPLIANCE

15.1 Regulatory Certifications

IEC 60947-2	Certified / Compliant		
CE	European Conformity	2014/30/EU (EMC), 2014/35/EU (LVD), 2014/68/EU (PED)	Full compliance
UL 489	Certified / Compliant		
CSA	Canadian Standards	CSA C22.2 No. 142	Certified
CCC	China Compulsory	GB standards	Certified
RoHS	Hazardous Substances	2011/65/EU + 2015/863	Full compliance

15.2 Declaration of Conformity

We, ElectroPower Systems, declare under our sole responsibility that the product CB400 Molded Case Circuit Breaker (Model: CB400-3P-250A) is in conformity with the provisions of the applicable EU Directives and harmonized standards listed above. A copy of the full EU Declaration of Conformity is available on request.

16 QUALITY ASSURANCE & TESTING

16.1 Quality Management System

ElectroPower Systems maintains a comprehensive quality management system certified to ISO 9001:2015. All products are manufactured under controlled conditions with full traceability of materials, processes, and test results.

16.2 Factory Acceptance Testing

Every device undergoes the following factory acceptance tests before shipment:

Test	Method	Acceptance Criteria	Record
Visual inspection	100% visual	No defects	Inspection report
Dimensional check	Gauge / CMM	Per drawing tolerances	Dimensional report
Pressure test	Hydrostatic 1.5x	No leaks, no deformation	Test certificate
Electrical safety	Hi-pot 1500 V AC	< 1 mA leakage	Test certificate
Insulation resistance	500 V DC	> 100 MOhm	Test certificate
Calibration	5-point up/down	Within accuracy spec	Calibration certificate
Functional test	Full operating cycle	All functions verified	Test report
Communication test	Protocol verification	Correct response	Test report
Burn-in	24-hour power-on	Stable output	Data log

16.3 Traceability

Each device is assigned a unique serial number that enables complete traceability from raw materials through manufacturing, testing, and delivery. All test records are maintained for a minimum of 10 years.

17 ENVIRONMENTAL TESTING RESULTS

17.1 Type Test Results

Test	Standard	Level / Duration	Result	Notes
Temperature cycling	IEC 60068-2-14	-40 to +85 C, 100 cycles	PASS	No degradation
Dry heat	IEC 60068-2-2	+85 C, 96 hours	PASS	Within spec
Cold	IEC 60068-2-1	-40 C, 96 hours	PASS	Within spec
Humidity (steady)	IEC 60068-2-78	40 C / 93% RH, 56 days	PASS	No corrosion
Humidity (cyclic)	IEC 60068-2-30	25-55 C, 6 cycles	PASS	No condensation damage
Vibration (sinusoidal)	IEC 60068-2-6	10-2000 Hz, 20g	PASS	No resonance shift
Vibration (random)	IEC 60068-2-64	10-2000 Hz, 10 grms	PASS	Functional during test
Shock	IEC 60068-2-27	100g, 6 ms, 3 axes	PASS	No damage
Salt spray	IEC 60068-2-52	96 hours, 5% NaCl	PASS	No corrosion on SS parts
EMC immunity	EN 61326-1	Industrial environment	PASS	Performance A
EMC emissions	EN 61326-1	Class A limits	PASS	Below limits
ESD	IEC 61000-4-2	8 kV contact, 15 kV air	PASS	Performance A
Surge	IEC 61000-4-5	2 kV line, 1 kV I/O	PASS	Performance B
IP rating	IEC 60529	IP67	PASS	Verified at factory

17.2 MTBF Calculation

The calculated Mean Time Between Failures (MTBF) for the CB400-3P-250A is:

Condition	MTBF (hours)	MTBF (years)	Method
Ground Benign (GB)	850,000	97	Telcordia SR-332
Ground Fixed (GF)	425,000	48.5	Telcordia SR-332
Ground Mobile (GM)	210,000	24	MIL-HDBK-217F
Naval Sheltered (NS)	180,000	20.5	MIL-HDBK-217F

18 APPLICATION NOTES

18.1 Typical Installation Configurations

The CB400-3P-250A can be installed in various configurations depending on the application requirements. The following examples illustrate common installation scenarios:

Configuration A: Direct Process Mount
The device is mounted directly on the process pipe or vessel using the integral process connection. This configuration provides the fastest response time and is recommended for most applications.

Configuration B: Remote Mount with Impulse Tubing
The device is mounted remotely from the process using impulse tubing. This configuration is used when direct mounting is not practical due to high temperature, vibration, or accessibility constraints. Keep impulse tubing as short as possible (< 3 m recommended) and slope continuously toward the process.

Configuration C: Manifold Mount
The device is mounted on a valve manifold for isolation and calibration purposes. This is the recommended configuration for critical process measurements and safety instrumented systems (SIS).

18.2 Best Practices

- Always install the device with the recommended orientation to minimize errors
- Use impulse tubing of the same material as the process piping
- Install drip legs or condensate pots for steam applications
- Use diaphragm seals for corrosive, viscous, or sanitary applications
- Consider using remote seals for high-temperature applications (> 120 C)
- Calibrate the device with the diaphragm seal attached (as a system)
- Document the installation configuration for future reference
- Include the device in the plant preventive maintenance program

18.3 Industry-Specific Guidelines

Industry	Key Considerations	Recommended Options	Relevant Standards
Oil & Gas	Hazardous area, NACE	ATEX, SIL, HC276	API, ISA, NACE MR0175
Chemical	Corrosive media	HC276/Titanium, PTFE seals	DIN, EN, NAMUR
Food & Beverage	Hygiene, clean-in-place	3-A, FDA, Ra < 0.8 um	EHEDG, 3-A SSI
Pharma	Validation, traceability	3.1 cert, NIST cal	GAMP 5, 21 CFR Part 11
Water/Wastewater	Outdoor, submersible	IP68, PUR cable	MCERTS, NSF 61
Power	High temp, SIL	HT version, SIL 2/3	IEC 61511, ASME

19 WARRANTY & SUPPORT

19.1 Standard Warranty

ElectroPower Systems warrants this product to be free from defects in materials and workmanship for a period of three (3) years from the date of shipment from the factory. During the warranty period, ElectroPower Systems will repair or replace, at its sole discretion, any product that is found to be defective under normal use and service conditions.

19.2 Warranty Exclusions

This warranty does not cover damage resulting from:

- Misuse, abuse, neglect, or unauthorized modification of the product
- Installation or operation outside the specified ratings and conditions
- Damage caused by process media incompatibility with wetted materials
- Normal wear of consumable components (seals, gaskets, O-rings)
- Damage during shipping (claims must be filed with the carrier)
- Use of non-approved spare parts or accessories
- Failure to follow the installation and maintenance instructions in this document
- Acts of nature, including but not limited to lightning, flooding, or earthquake

19.3 Technical Support

For technical support, warranty claims, or spare parts orders, contact ElectroPower Systems:

- Email: support@electropowersystems.com
- Phone: +49 (0) 123 456-789 (Europe) / +1-800-555-0199 (North America)
- Web: www.electropowersystems.com/support
- 24/7 Emergency Hotline: +49 (0) 123 456-999

19.4 Return Authorization

All products being returned for warranty repair or evaluation must have a Return Material Authorization (RMA) number issued by the factory. Contact the support team to obtain an RMA number before shipping any product. Products returned without an RMA number may be refused or delayed.

20 REVISION HISTORY

Rev	Date	Author	Description of Changes	Approved
A	2023-06-15	Engineering	Initial release	QA Manager
A1	2023-09-20	Engineering	Added IO-Link communication option	QA Manager
A2	2024-01-10	Engineering	Updated EMC test results per EN 61326-1:2021	QA Manager
A3	2024-03-05	Applications	Added application notes for food & beverage	QA Manager
B	2025-01-15	Engineering	Major revision: Added SIL certification, updated performance data, added new orderi	Director of Engineering

Document Information

Document Number	DS-CB400-EN-RevB
Current Revision	B
Effective Date	January 15, 2025
Classification	Public
Language	English
Total Pages	28

End of Document - CB400 Molded Case Circuit Breaker Technical Datasheet & Installation Manual